

Name: _____

Class: 2ma2____ / 2IPma2____



NATIONAL JUNIOR COLLEGE
2009 SH2 Preliminary Examination

MATHEMATICS

Higher 2

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Sheet

9740/01

Paper 1

3 hours

04 September 2009

1430 – 1730 hours

INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group, on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **ALL** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in the brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of 7 printed pages.

- 1** A graphic calculator is not to be used in answering this question.

Solve the inequality $x \leq \frac{15}{x+2}$. Hence, solve the inequality $\frac{1}{x^2} \geq \frac{x^2+2}{15}$. [5]

- 2** The function f is defined by

$$f : x \mapsto (2-x)(2+x) \quad x \in \mathbb{R}, x \geq 3.$$

- (i) Find the inverse of f . State clearly its domain. [3]

- (ii) A linear function g has domain $(-\infty, a]$. Given that gf exists such that $gf(x) = 2x^2 + 3$, find g , and state the range of possible values for a . [3]

- 3** The points A , B and C have position vectors $3\mathbf{i} - \mathbf{j}$, $5\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $7\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ respectively.

- (i) Calculate the exact area of the triangle ABC . [2]

The point N is on CB produced such that $BN : BC = \lambda : 1$.

- (ii) Find $\overrightarrow{AN}$ and determine the value of λ such that AN is perpendicular to BC . [4]

- 4** The complex number z satisfies both $|iz + 1| \leq 2$ and $\frac{3\pi}{4} \leq \arg(z - 1) \leq \pi$.

- (i) Illustrate the locus of the points representing z in an Argand diagram. [3]

- (ii) Calculate the least possible value of $|z - 3i|$. [2]

- (iii) Sketch the locus of the points representing w that satisfies both $|iw^* + 1| \leq 2$ and

$$\frac{3\pi}{4} \leq \arg(w^* - 1) \leq \pi \text{ on the same Argand diagram in (i). [1]}$$

- 5** By using the substitution $y = vx$, show that the differential equation

$$x \frac{dy}{dx} - y = x(x - y) \text{ where } x > 0$$

reduces to $\frac{dv}{dx} = 1 - v$. [2]

Find the general solution of the differential equation $x \frac{dy}{dx} - y = x(x - y)$, expressing y in terms of x . [4]

- 6** By considering $u_r - u_{r+1}$ where $u_r = \frac{1}{r(r+1)}$, show that

$$\sum_{r=1}^N \frac{1}{r(r+1)(r+2)} = \frac{1}{4} - \frac{1}{2(N+1)(N+2)}. \quad [3]$$

(i) Deduce $\sum_{r=2}^N \frac{1}{r(r^2-1)}$. [2]

(ii) Find the value of $\frac{1}{10(10^2-1)} + \frac{1}{(11)(11^2-1)} + \frac{1}{12(12^2-1)} + \dots$. [2]

- 7 (a)** In a triangle ABC , $AB = 3$ cm and $AC = 2$ cm. If angle BAC is increasing at a constant rate of 0.1 radians per second, find the rate of increase of the length BC at the instant where $BAC = \frac{\pi}{3}$ radians. [4]

- (b)** Suppose p is a positive constant. Find the coordinates of the points on the curve

$$x^2 - xy = p^2 + y^2$$

at which the normal is parallel to the x -axis. Leave your answers in terms of p . [4]

- 8 (a)** Write down the constants A and B such that, for all values of x ,

$$x = A(2x + 4) + B.$$

Find $\int \frac{x}{x^2 + 4x + 6} dx$. [5]

(b) (i) Show that $\frac{d}{dx} \left(\frac{1}{\sqrt{x^2 - 1}} \right) = -\frac{x}{(x^2 - 1)^{\frac{3}{2}}}$. [1]

(ii) By using the substitution $x = \frac{1}{y}$, find $\int \frac{1}{x\sqrt{x^2 - 1}} dx$. [3]

(iii) Find $\int x(x^2 - 1)^{-\frac{3}{2}} \ln x dx$. [2]

- 9 (a)** Given that $z = -\sqrt{3} + i$, express z in the form $re^{i\theta}$ where r is real and $-\pi < \theta \leq \pi$. Hence find exactly the values of a and b which satisfy the equation $(iz)^4 = e^{a+ib}$, where $a \in \mathbf{R}$, and $-\pi < b \leq \pi$. [4]

- (b)** Solve $w^3 + 8 = 0$ where w is a complex number. Leave your answers in the exact form $x + iy$ where x and y are real. [3]

Deduce the roots of $w^4 + w^3 + 8w + 8 = 0$ [2]

Determine the roots of z such that $z^4 - iz^3 + 8iz + 8 = 0$. [2]

- 10** The points A and B have position vectors $-\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + 7\mathbf{k}$ respectively.

The plane Π_1 is given by $\mathbf{r} \cdot (\mathbf{i} - \mathbf{j}) = 5$.

- (i) Another plane Π_2 is perpendicular to Π_1 and contains the points A and B . Find the equation of Π_2 in the form $\mathbf{r} \cdot \mathbf{n} = d$. [3]
- (ii) Find $\overrightarrow{ON}$, where N is the foot of perpendicular from the point A to the plane Π_1 . [3]
- (iii) Deduce the equation of the line of intersection of the planes Π_1 and Π_2 . [2]

The plane Π_3 has vector equation $\mathbf{r} \cdot (4\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = D$, where D is a positive constant.

- (iv) Determine if the planes Π_1 , Π_2 and Π_3 meet at common point or a common line. [2]
- (v) Given that the shortest distance of the point B from the plane Π_3 is $\sqrt{21}$, find the value of D . [3]

- 11** A sequence A has first term $\ln(2x)$ and the sum of the first n terms is $\frac{n}{2} \ln(2^{n+1} x^2)$ where x is an unknown constant.

- (i) Show that the n^{th} term of the sequence is $\ln(2^n x)$. Prove that A is an arithmetic progression. [4]

The first two terms of A form two consecutive terms of a geometric progression G respectively. Write down the common ratio of G . [1]

- (ii) Determine the range of values of x such that G is convergent. [2]
- (iii) Taking $x = 0.2$, find the first term of G such that its sum to infinity is half of the sum of the first 40 terms of A . [2]

12 The curve C is the graph of $f(x) = \frac{k(x^2 - a^2)}{x - 5}$, where k and a are real numbers with $a > 0$

and $k < 0$. C has a minimum point at $x = 1$.

(i) Show that $a = 3$. [2]

Leaving your answers in terms of k ,

(ii) obtain the equations of the asymptotes of C . [3]

(iii) determine the coordinates of the turning points of C . [3]

(iv) sketch the curve C and the graph of $y = f'(x)$ on two separate diagrams. Label your graphs clearly. [4]

End of Paper



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INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group and subject tutor's name in the spaces provided on the cover sheet and attach it on top of your answer paper.

Circle the questions you have attempted and arrange your answers in **NUMERICAL ORDER**.

Name: _____ Registration Number: _____

Subject Tutorial Group: 2ma2____ / 2IPma2____ Subject Tutor: _____

Question	Marks	Max. Marks	Question	Marks	Max. Marks
1		5	7		8
2		6	8		11
3		6	9		11
4		6	10		13
5		6	11		9
6		7	12		12
Presentation: -1 / -2					
Total		100	Grade		